

# Architecture Design Report

This document presents the architectural design of the system, including its context, logical structure, runtime behavior, and deployment details.

## 1. System Requirements

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System Name: ResuFit: Precision CV Matching System

Architecture Style: Microservices

### Functional Requirements

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- Submit CV
- View and Edit CV
- Answer Screening Questions
- Get Rating for CV

### Non-Functional Requirements

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#### - Performance

Description: The system **MUST** respond within 2 seconds to user input.

#### - Reliability

Description: The system **SHALL** be available 99.99% of the time.

#### - Security

Description: The system **MUST** protect user data and **MUST NOT** store sensitive information.

#### - Privacy

Description: The system **MUST NOT** collect or store user data without explicit consent.

#### - Scalability

Description: The system **SHOULD** be able to handle 1000 concurrent users.

#### - Usability

Description: The system **MUST** be user-friendly and **MUST** provide clear instructions.

#### - Maintainability

Description: The system **SHOULD** be easy to modify and update.

## - Compatibility

Description: The system MUST be compatible with major web browsers.

## - Integration

Description: The system MUST integrate with HR module from Odoo ERP software.

## 2. Formal Architecture Specification (ACME ADL)

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The following section provides a formal architectural specification using ACME ADL, capturing components, co

```
System ResuFit:PrecisionCVMatchingSystem {
  Property view : String = "runtime";
  Property architectureStyle : String = "Microservices";
  Property source : String = "AI";
  Component CVParser = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Parse and analyze submitted CVs";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
  }
  Component CVEditor = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Allow applicants to view and edit their parsed CVs";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
  }
  Component ScreeningQuestionManager = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Manage and present screening questions defined by expert rec
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
  }
  Component RatingGenerator = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Generate a percentage-based result for each applicant's CV";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
  }
  Attachment CVParser.out to CVEditor.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = true;
    Property backpressure : String = "bounded";
  }
  Attachment CVEditor.out to ScreeningQuestionManager.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = false;
    Property backpressure : String = "bounded";
  }
  Attachment ScreeningQuestionManager.out to RatingGenerator.in {
```

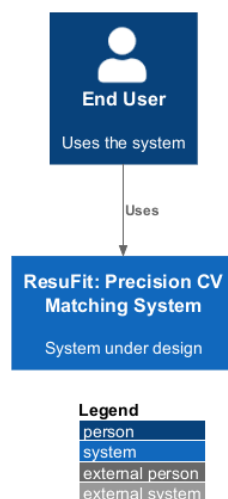
```

    Property delivery : String = "at-least-once";
    Property ordering : Boolean = true;
    Property backpressure : String = "bounded";
}
Attachment CVParser.out to RatingGenerator.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = true;
    Property backpressure : String = "bounded";
}
Property resilience_pattern : String = "Retry, CircuitBreaker";
Property delivery : String = "at-least-once";
Property ordering : Boolean = false;
Property authentication : String = "service-to-service";
Property authorization : String = "RBAC";
}

```

### 3. Context View

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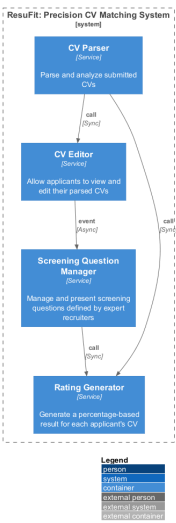


#### 3.1 Data Context View (Level 0 DFD)

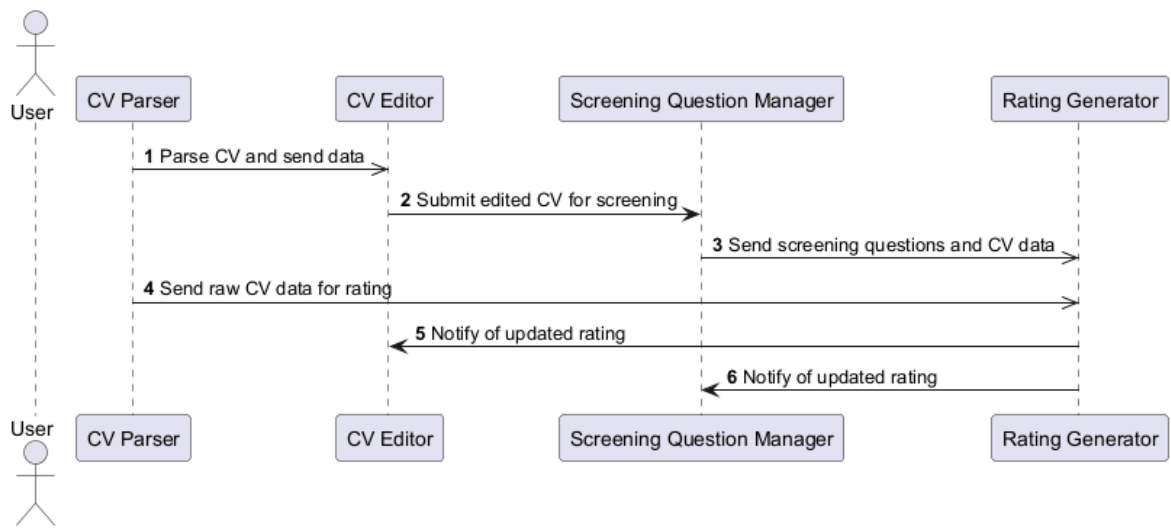
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## 4. Logical View (C4 Container Diagram)



## 5. Process View (Runtime Interaction)



## 6. Physical View (Deployment Diagram)

